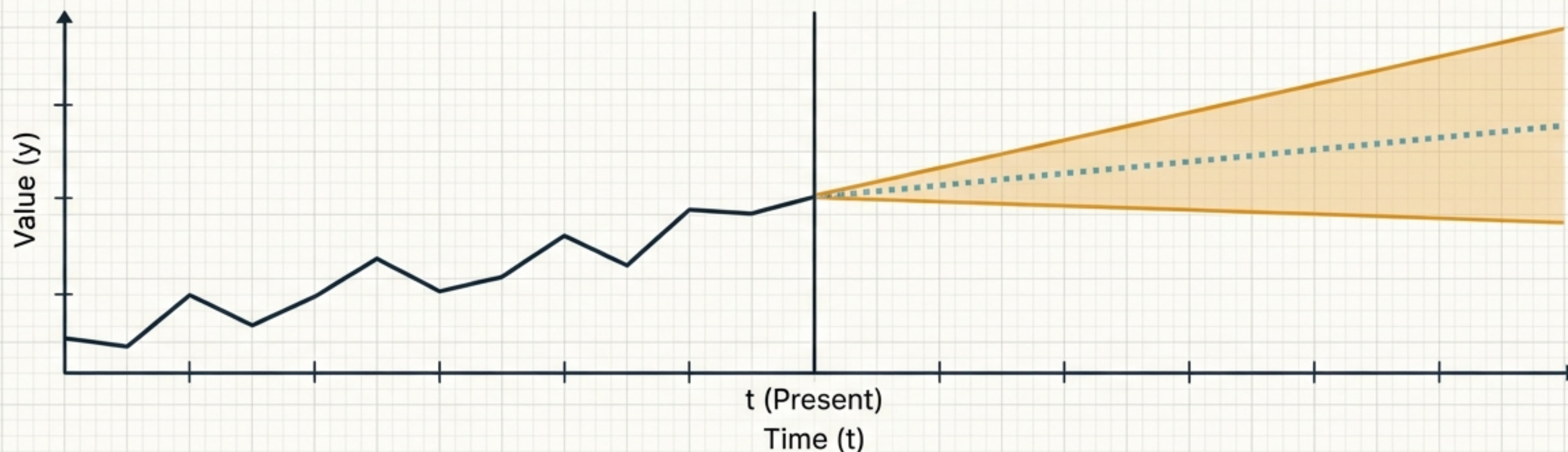


Mastering Time Series Analysis & Statistical Diagnostics

A Comprehensive Blueprint for Forecasting, Stationarity, and Hypothesis Testing



1. Core Anatomy



2. Statistical Engine



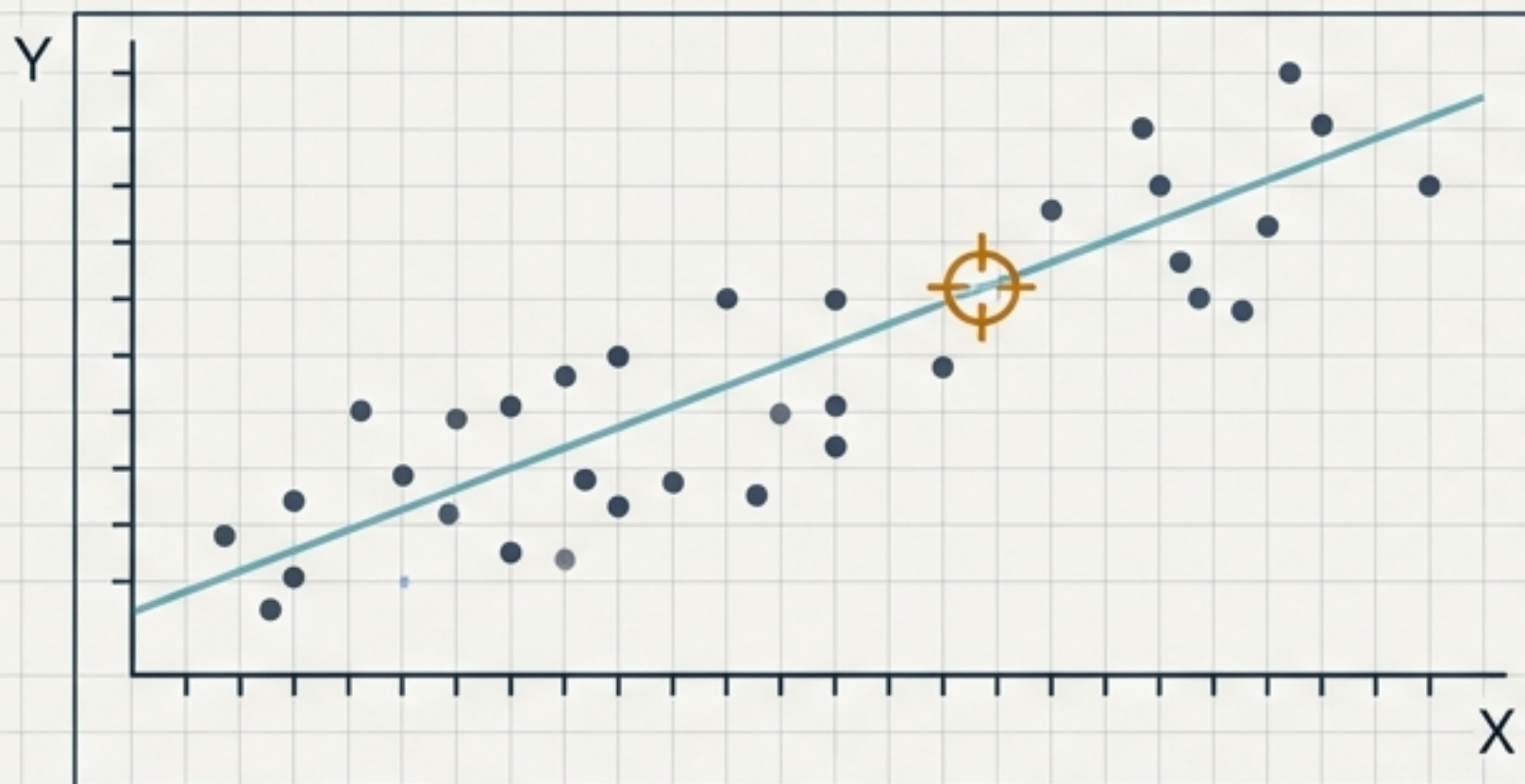
3. The Lab Tests



4. Forecasting



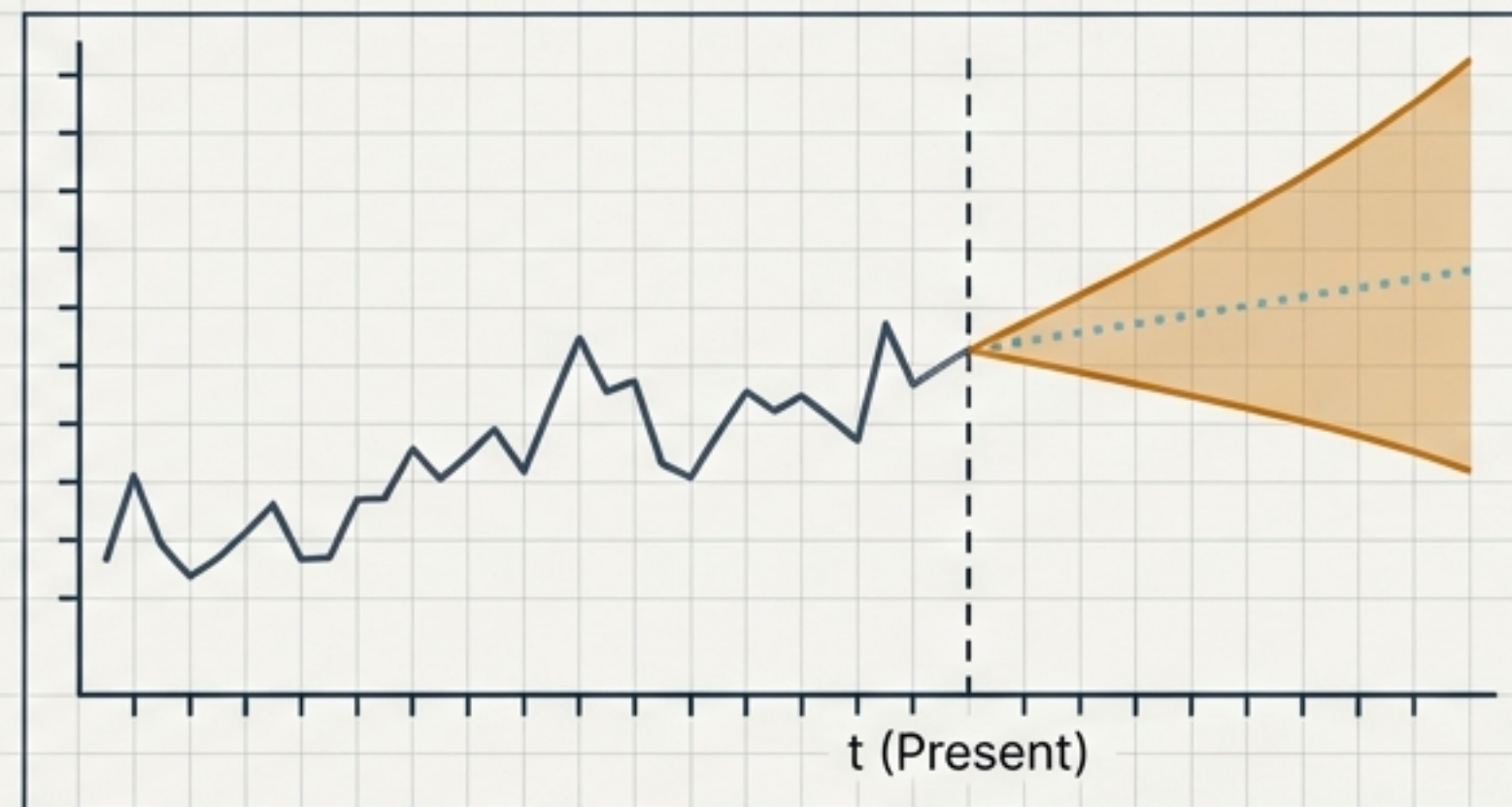
The Problem of Extrapolation: Time Series vs. Cross-Sectional



Interpolation (Cross-Sectional Data)

- Target: Fill the gaps.
- Mechanism: Predicts unknown values within the established range of past data.
- Assumption: Predictors are completely independent.

Equation: $y = mx + c$



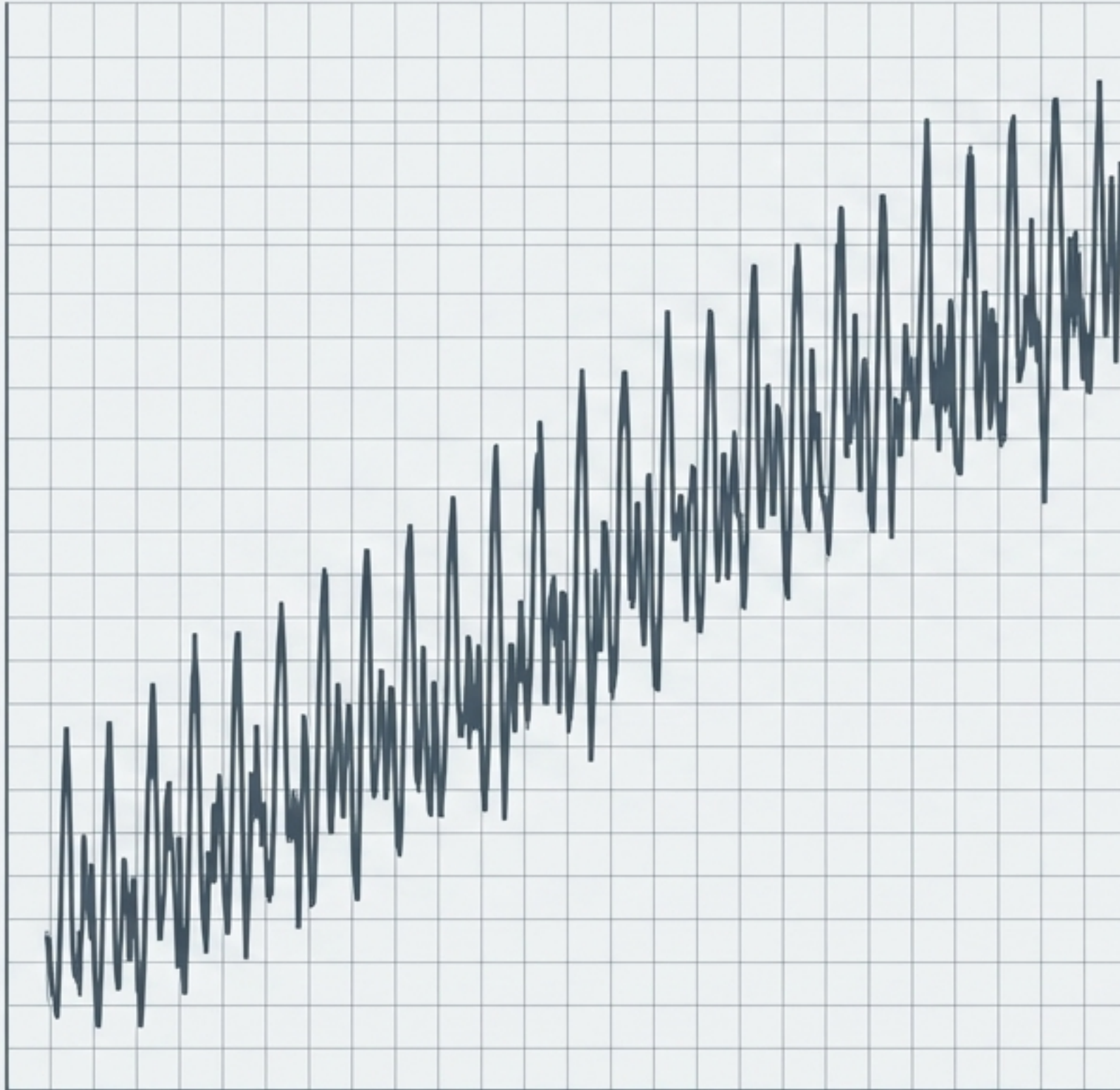
Extrapolation (Time Series Data)

- Target: Forecast the unknown.
- Mechanism: Predicts future values ($t+1$, $t+2$) based entirely on previous lags ($t-1$, $t-2$).
- Assumption: Data points are dependent on time.

KEY INTUITION: In extrapolation, error compounds over time. Forecasting a stationary series is required to limit this compounding error.

Signal Extraction: The Anatomy of a Time Series

Raw Observed Signal



1. Trend Component



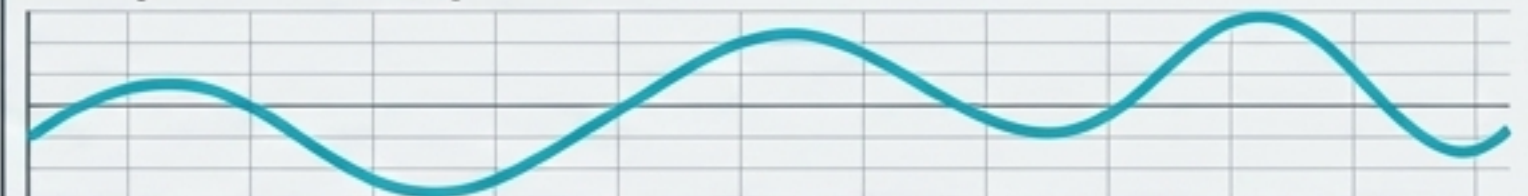
The general, long-term direction of the data. Does not repeat.

2. Seasonality Component



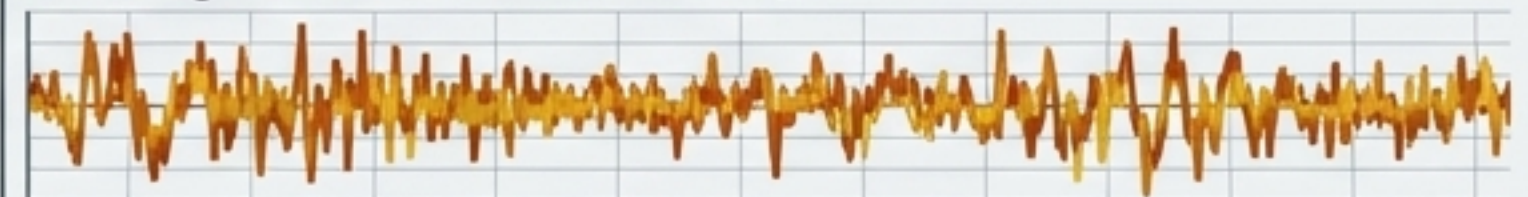
Fixed and known frequency. Tied to the calendar or clock (e.g., December retail spikes).

3. Cyclical Component



Ups and downs with no set repetition or fixed calendar frequency (e.g., 6-10 year economic boom/bust cycles).

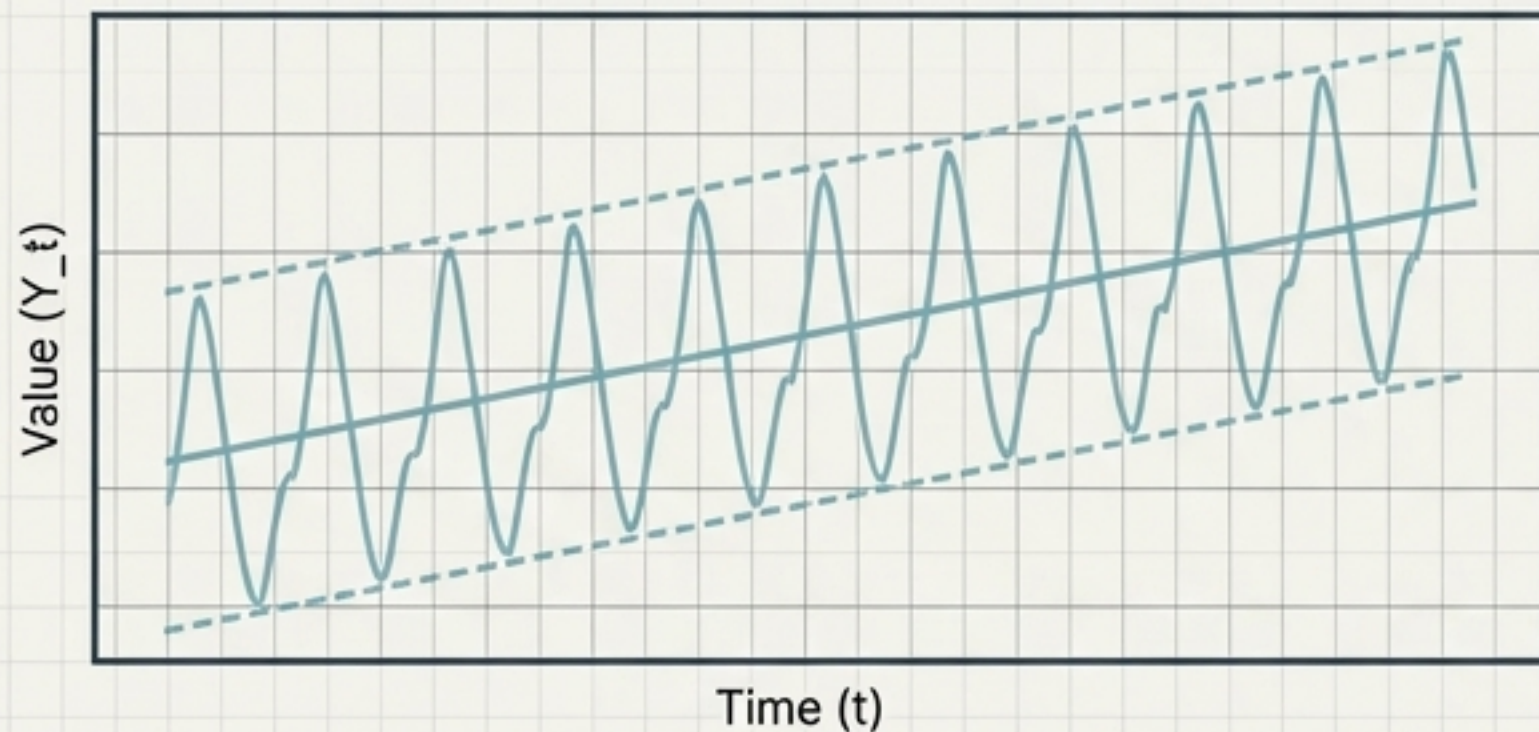
4. Irregular Variation (Noise)



Unpredictable fluctuations remaining after all pattern components are extracted.
The error term.

Decomposition Mechanics: Additive vs. Multiplicative

The Additive Model

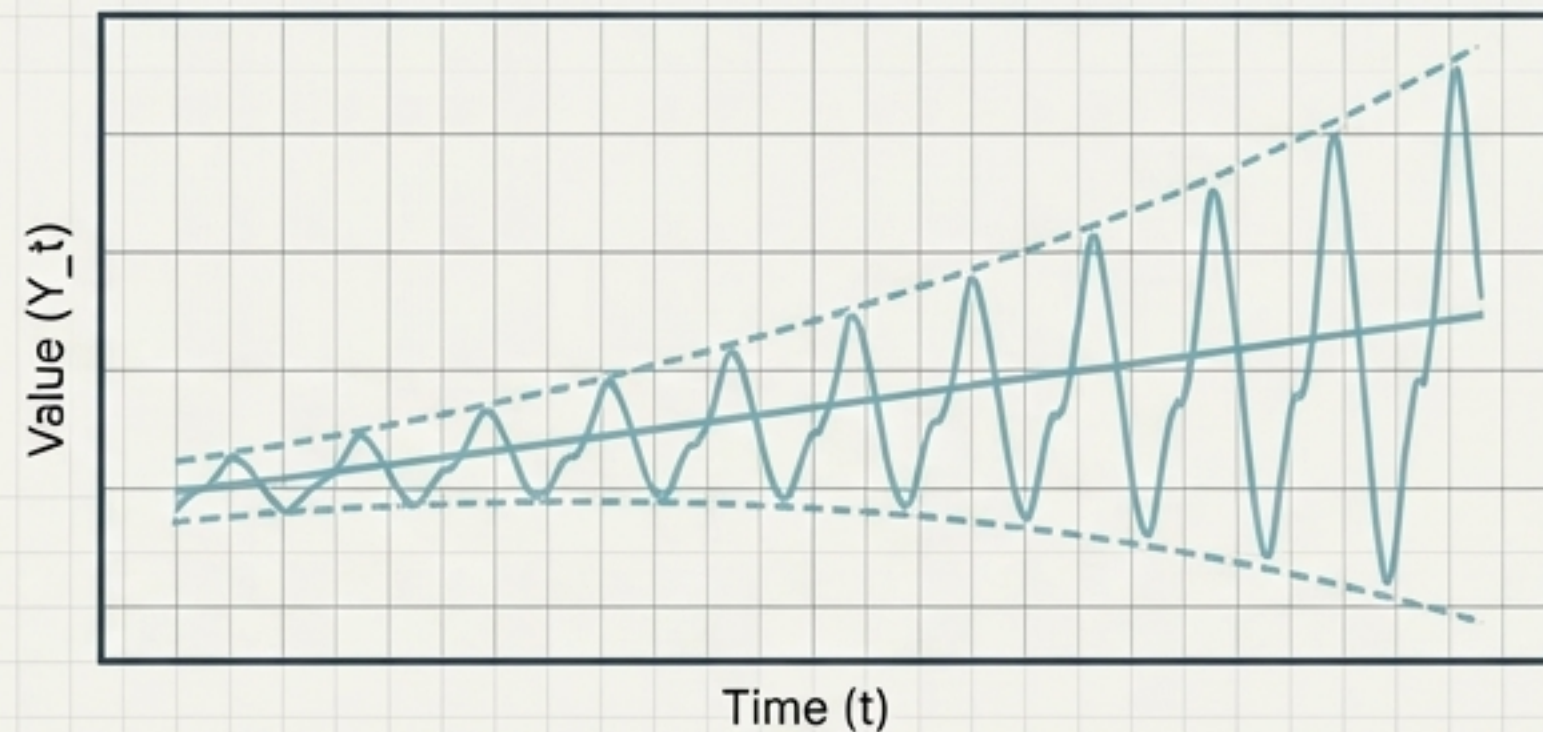


$$Y_t = T_t + S_t + C_t + I_t$$

Value = Base + Trend + Seasonality + Error

Condition: Use when the variance and magnitude of seasonality stay strictly constant over time, regardless of the trend.

The Multiplicative Model



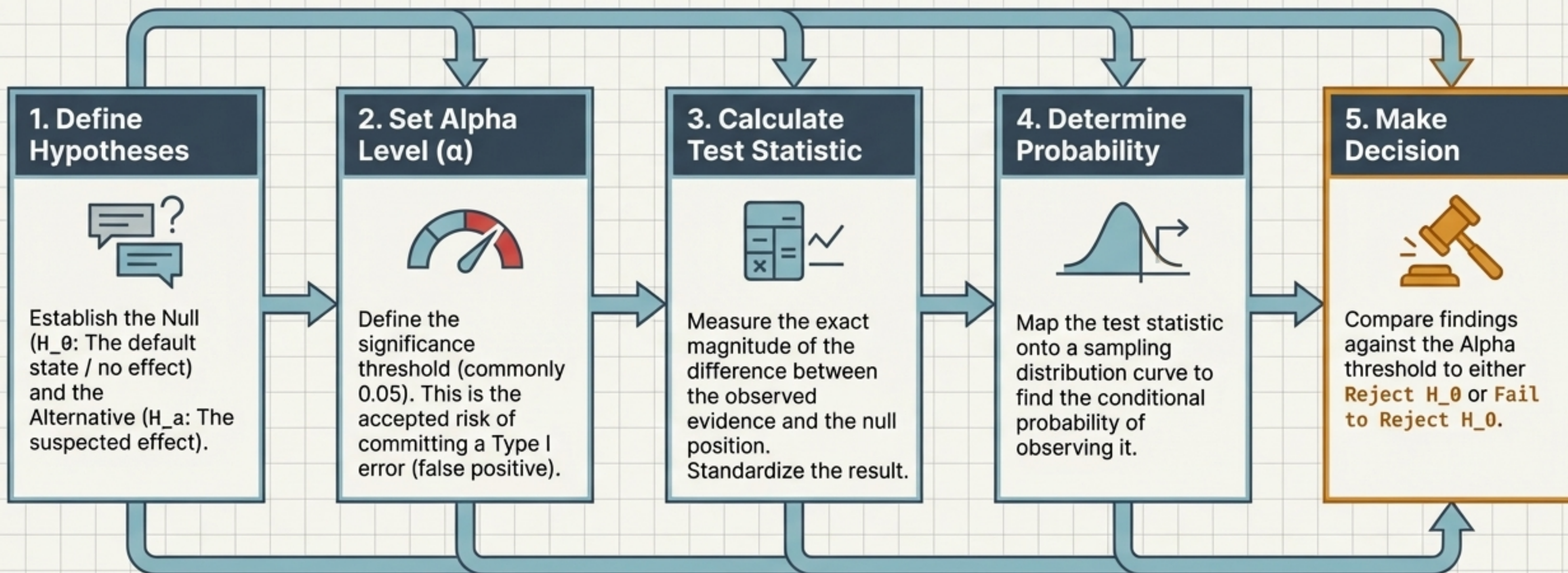
$$Y_t = T_t * S_t * C_t * I_t$$

Value = Base * Trend * Seasonality * Error

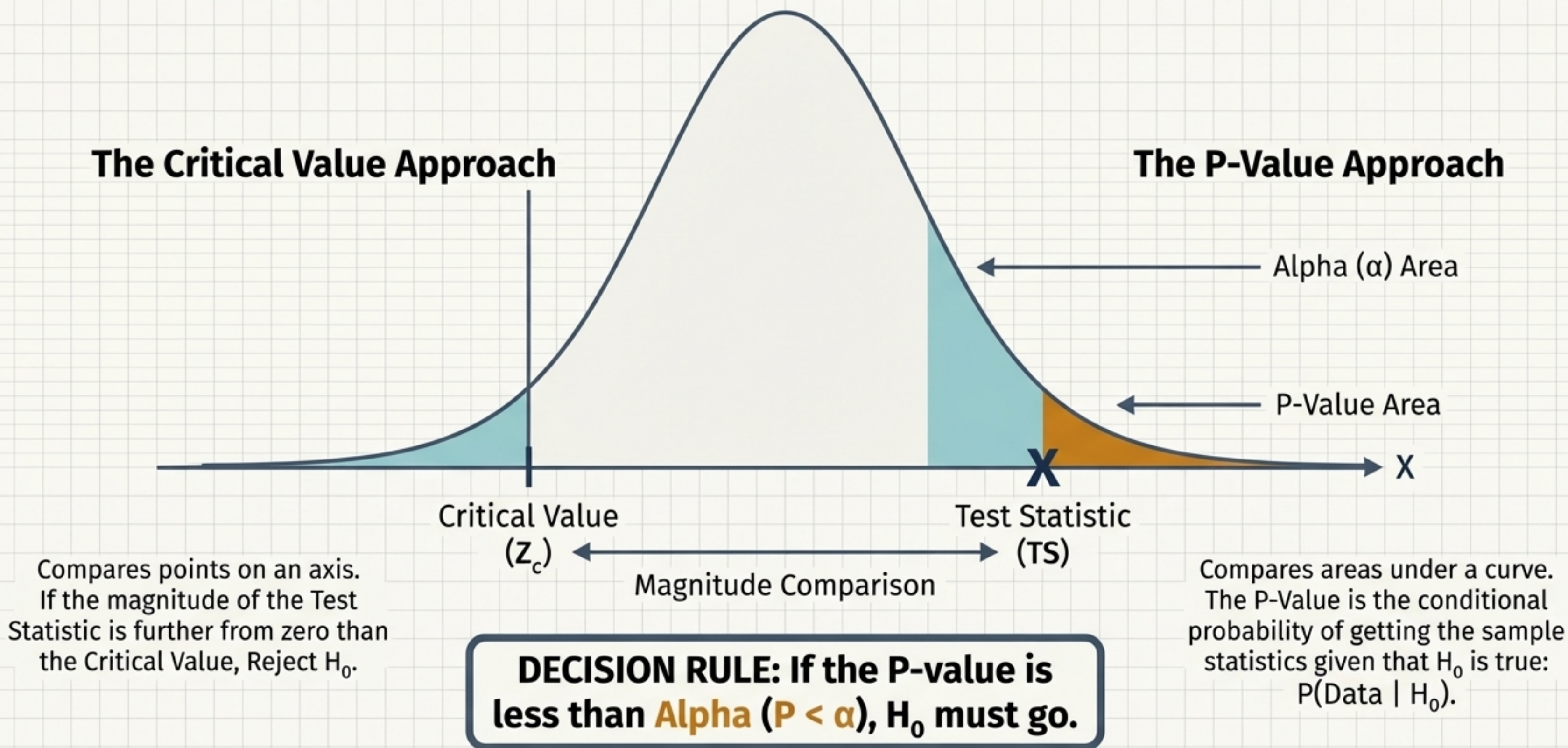
Condition: Use when the seasonal variation increases or decreases proportionally with the trend.

Transformation Fix: Apply a logarithmic transformation (log) to successfully convert a multiplicative series into an additive one.

Statistical Foundations: The Hypothesis Testing Engine

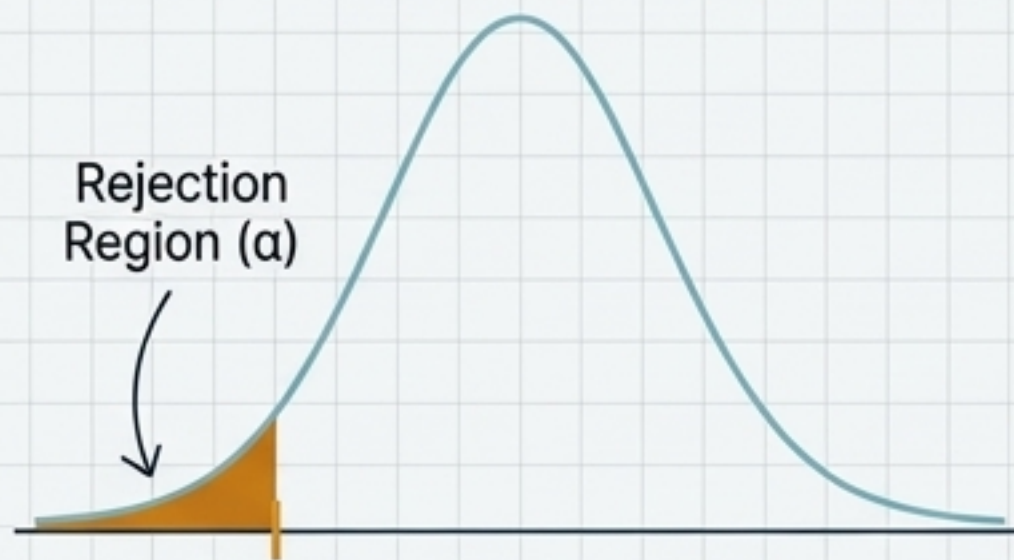


Step 5 Diagnostics: Critical Value vs. P-Value Approach



Tailing the Test: Directional Hypotheses

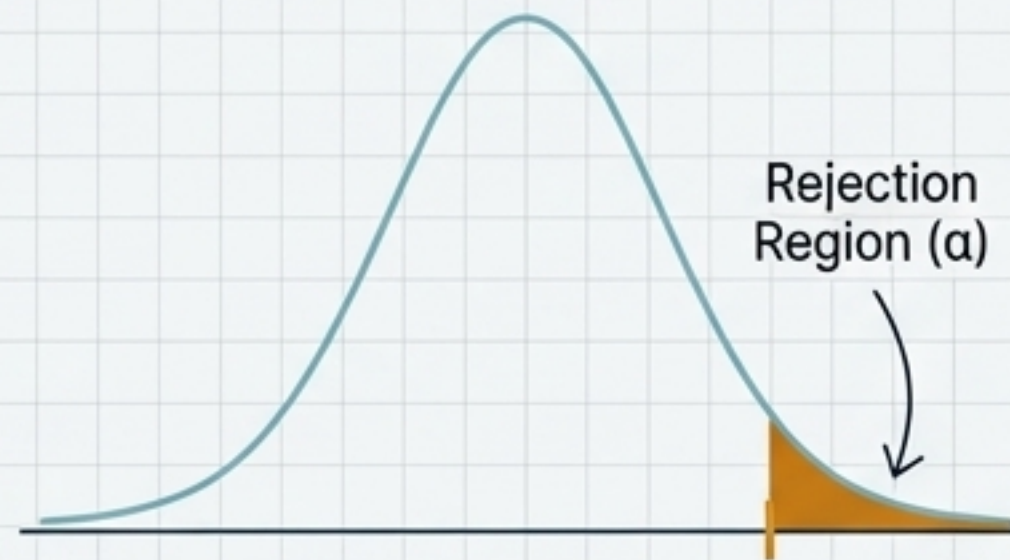
Left-Tailed Test



$$H_a: \mu < \text{Target}$$

Used when the Alternative Hypothesis states the mean is strictly less than a specific value. The entire Alpha area is concentrated on the left.

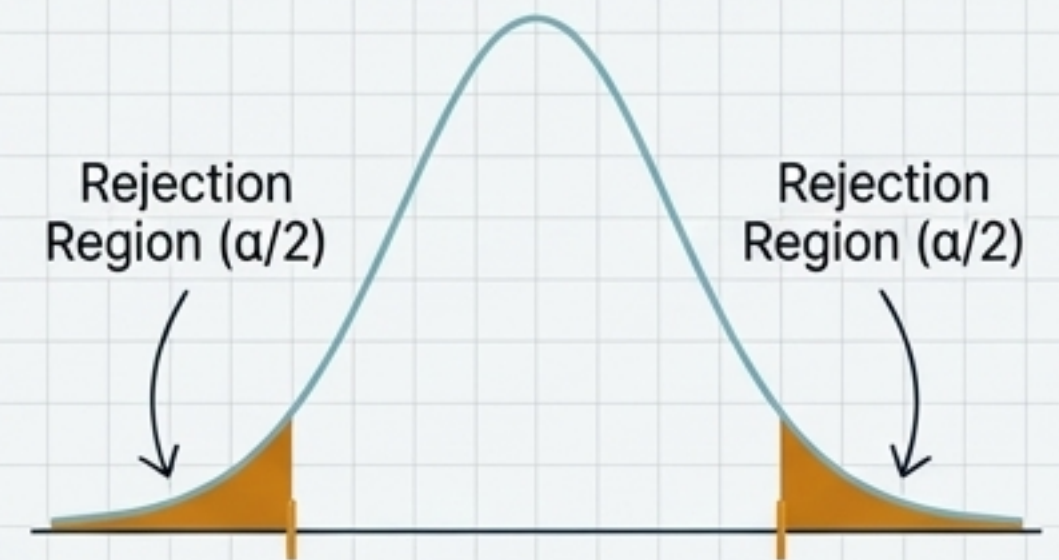
Right-Tailed Test



$$H_a: \mu > \text{Target}$$

Used when the Alternative Hypothesis states the mean is strictly greater than a specific value. The entire Alpha area is concentrated on the right.

Two-Tailed Test

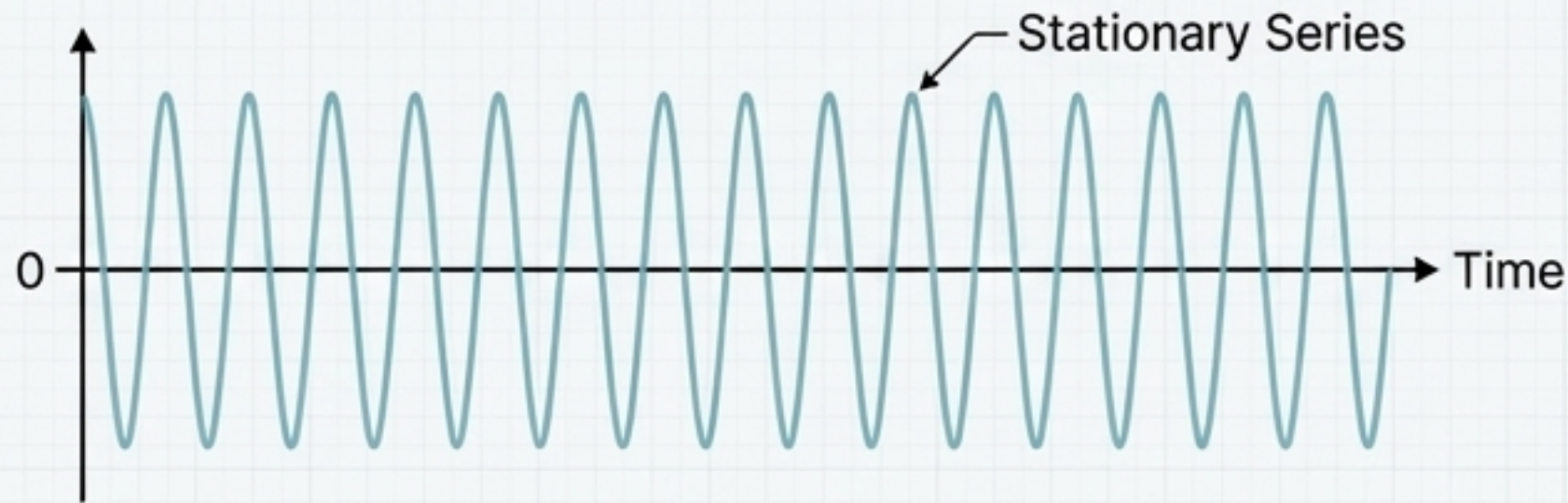


$$H_a: \mu \neq \text{Target}$$

Used when the Alternative Hypothesis states the mean is simply not equal to a target value. The Alpha area is split in half ($\alpha/2$) on both sides.

The Golden Rule: Stationarity & White Noise

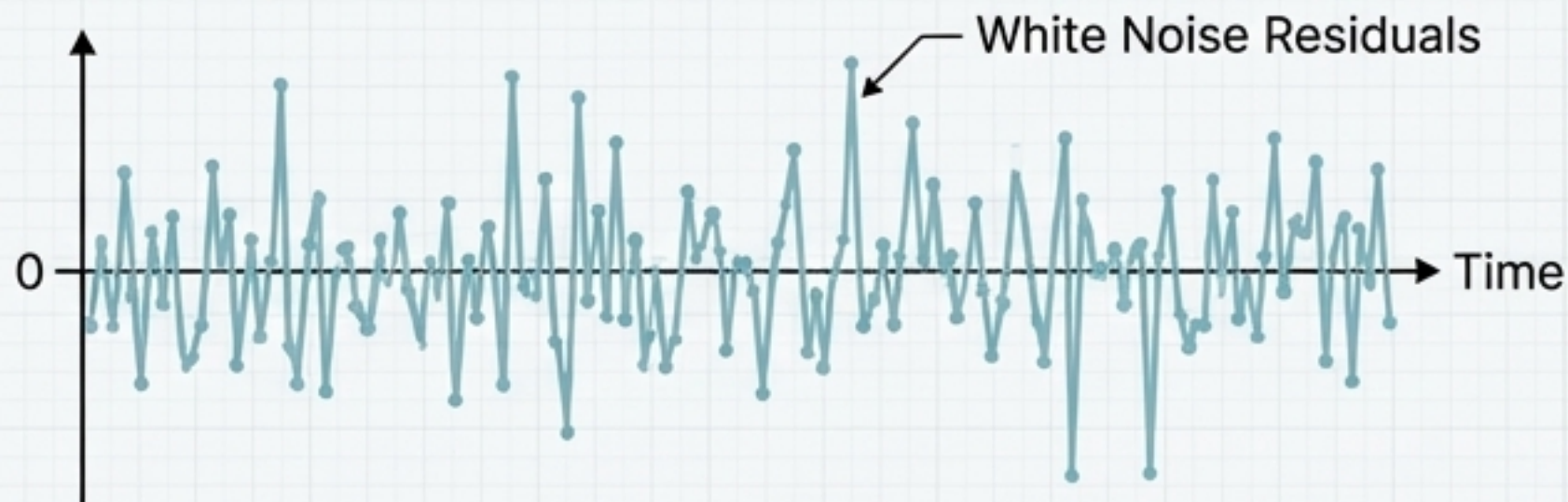
The Goal: Stationarity



Definition: A series whose statistical properties do not depend on the time at which the series is observed.

- **The 3 Constants:** Mean (μ), Variance (σ^2), and Autocorrelation are strictly constant over time.
- **Why it matters:** Autoregressive models are linear regressions using lags as predictors. Stationarity removes persistent autocorrelation, making predictors nearly independent.

The Ideal Residual: White Noise



Definition: A sequence of completely random, independent, and identically distributed (IID) numbers.

- **Properties:** Mean of exactly zero. No pattern.
- **Diagnostic Goal:** If a model's forecast errors (residuals) are white noise, the model has successfully extracted all possible signals. Forecasting is complete.

The Stationarity Diagnostic Matrix (Unit Root Tests)

Resolving the historical confusion of flipped hypotheses.

Test Name	Null Hypothesis (H_0)	Alternative Hypothesis (H_a)	Interpretation ($P < 0.05$)
ADF (Augmented Dickey-Fuller)	Non-Stationary (Has a unit root)	Stationary	Series is Stationary
KPSS (Kwiatkowski-Phillips-Schmidt-Shin)	Trend Stationary	Non-Stationary	 Series is Non-Stationary
PP (Phillips-Perron)	Non-Stationary (Integrated of order 1)	Stationary	Series is Stationary

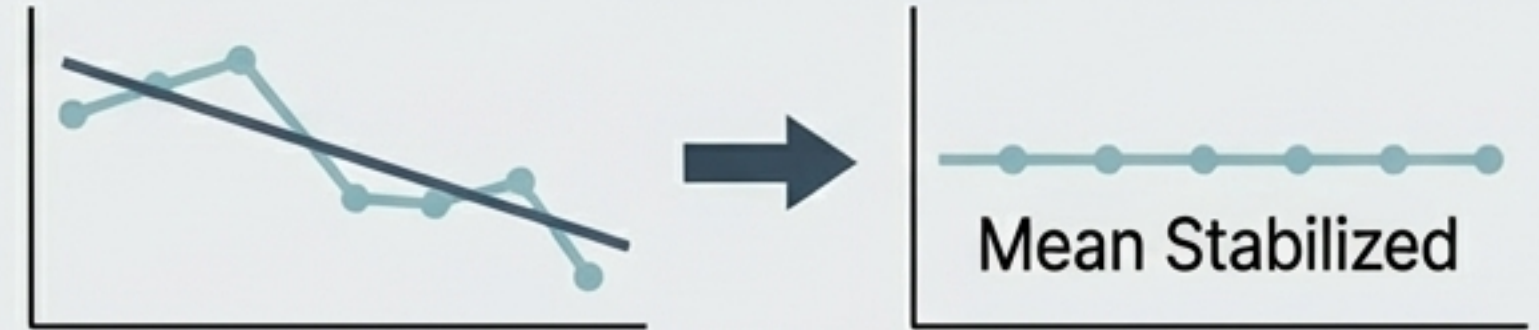


CRITICAL WARNING: The null hypothesis for the KPSS test is the exact opposite of the ADF test. Always cross-reference your P-value interpretations to avoid falsely identifying a series as stationary.

The Antidote: Making Series Stationary

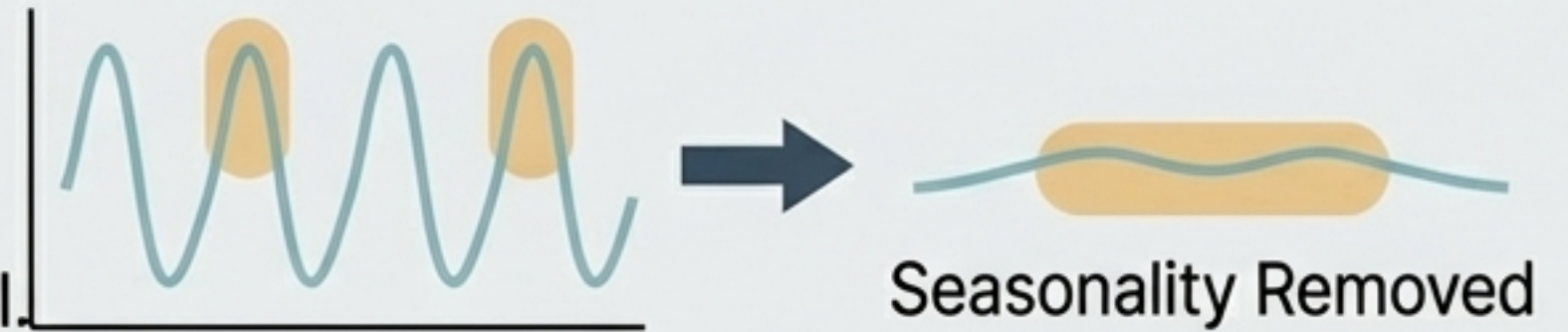
1. Detrending

Subtract the line of best fit (linear regression model with time steps as predictors) or subtract a moving average to stabilize the mean.



2. Deseasonalizing

Divide the data by the seasonal index. If ineffective, apply a **log transformation** to handle multiplicative variance, **deseasonalize**, then restore with an exponential.



3. Differencing (The Primary Cure)

Subtract the current value by the previous value to completely stabilize the mean.

First Difference: $\Delta Y_t = Y_t - Y_{t-1}$

Second Difference: $\Delta^2 Y_t = (Y_t - Y_{t-1}) - (Y_{t-1} - Y_{t-2})$



WARNING: Over-differencing produces wrong estimates and artificial patterns. Stop differencing as soon as the ADF test confirms stationarity.

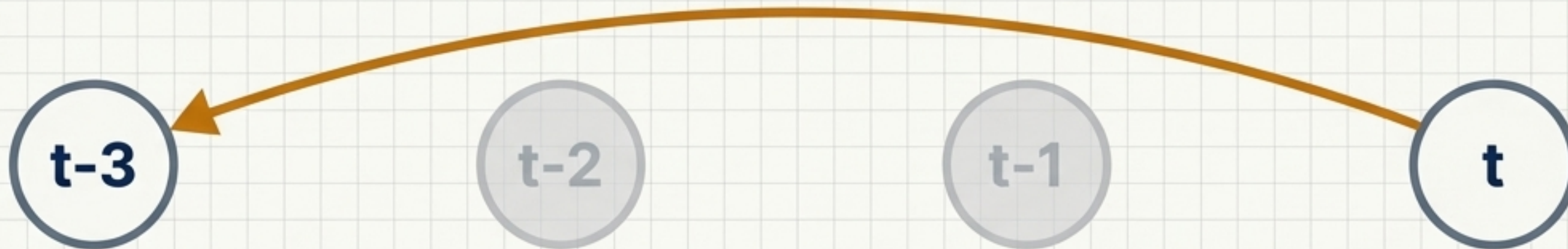
Time Series Diagnostics: Internal Memory (ACF vs. PACF)

ACF (Autocorrelation Function)



Correlation of a series with its own lags. Includes all intermediate indirect correlations. Excellent for spotting seasonality (spikes appear at exact multiples of the seasonal window).

PACF (Partial Autocorrelation Function)



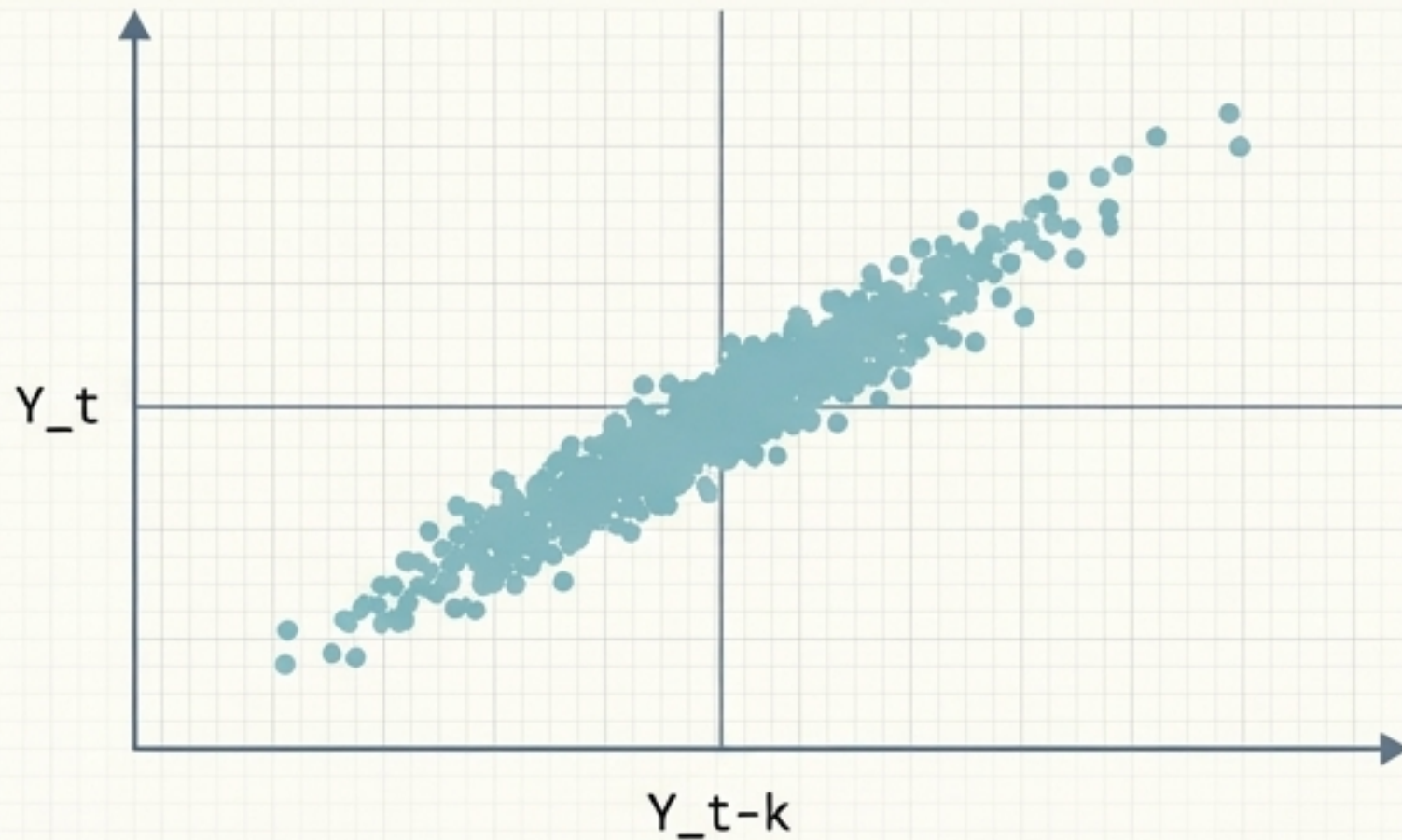
The PURE correlation of a series and its lag, excluding all correlation contributions from intermediate lags.

Math Intuition:

It is the coefficient α_k of the k -th lag in the autoregression equation.

Visualizing Relationships: Internal Memory vs. External Causes

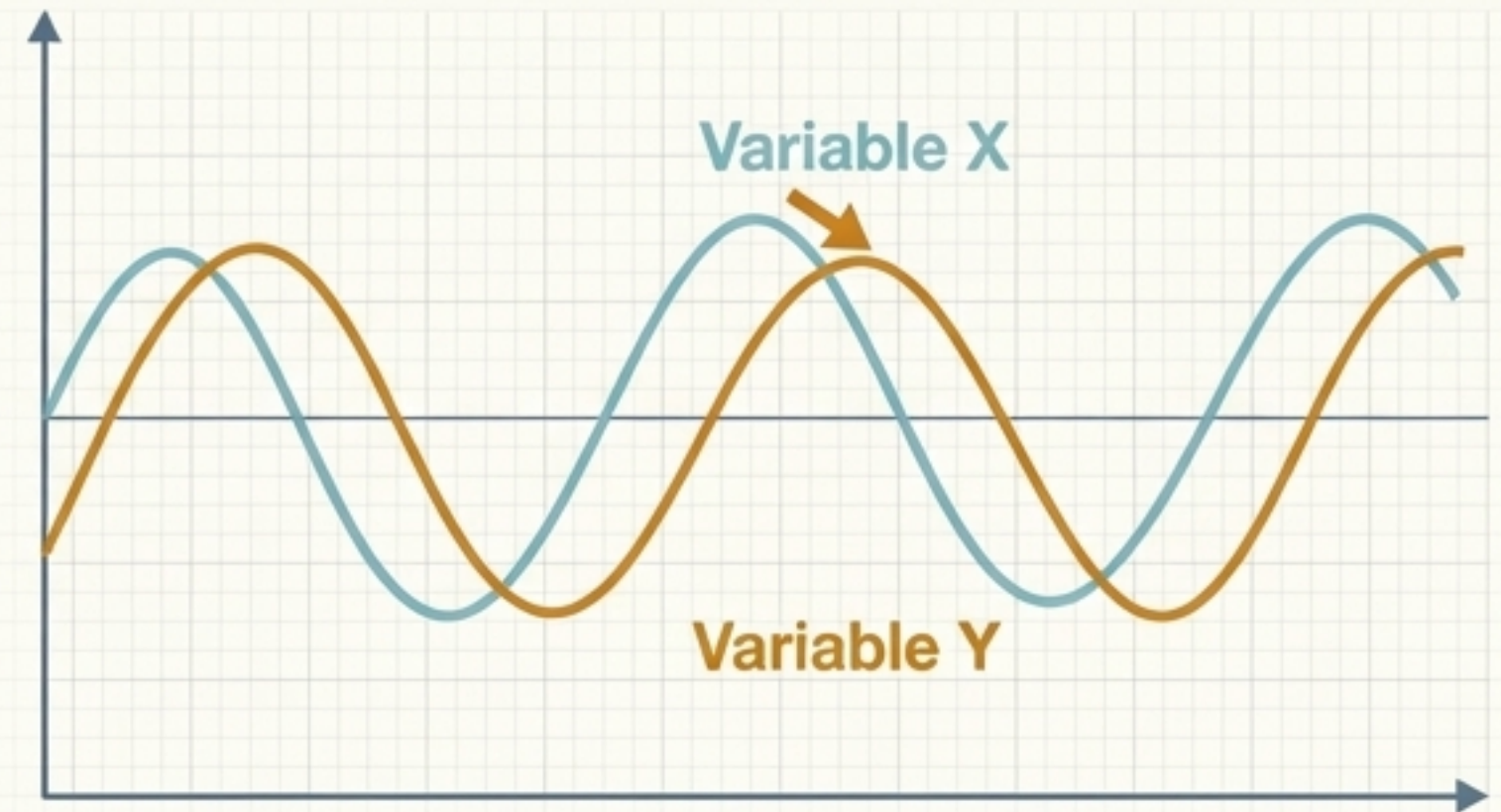
Lag Plots (Testing Self-Memory)



Concept: Maps the current observation against a previous lag.

Diagnosis: A strong linear diagonal pattern indicates high autocorrelation (the series has memory). A completely random, dispersed cloud indicates random white noise.

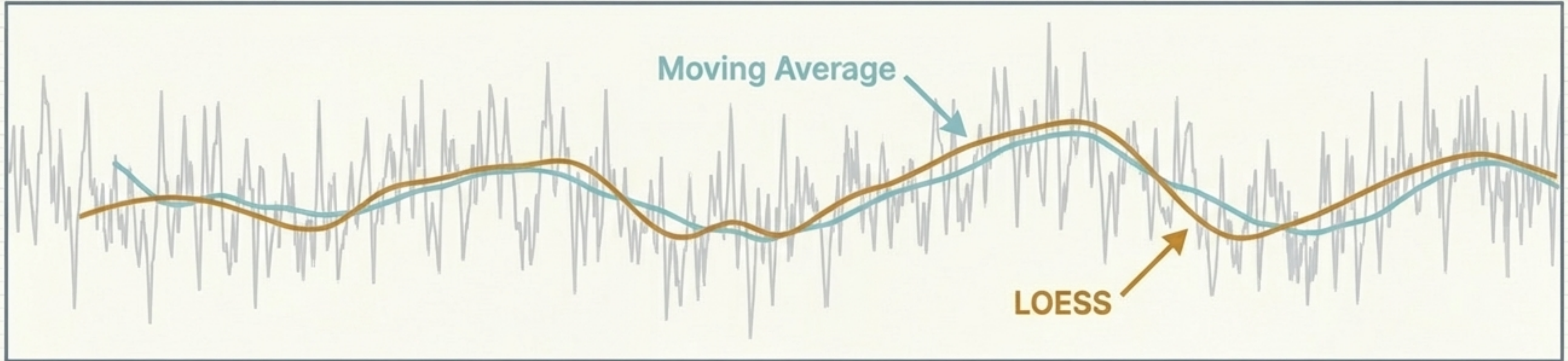
Granger Causality (Testing Exogenous Variables)



Concept: Tests if lagged values of exogenous variable X help predict Y better than Y's own lags alone.

Hypothesis Framework: Null (H_0) states that X does not Granger-cause Y. Reject H_0 to mathematically prove X is a useful external predictor.

Signal Extraction: Smoothing Techniques



Why Smooth?

To mechanically reduce noise, extract core features, and accurately visualize the true underlying trend without the distraction of erratic volatility.

Technique 1: Moving Average

The unweighted average of a rolling window of defined width. Moves step-by-step through the data.

Danger: Choosing a window size exactly equal to the seasonal duration will perfectly flatline and artificially nullify the seasonal effect (over-smoothing).

Technique 2: Localized Regression

LOESS / LOWESS fits multiple regressions in the immediate local neighborhood of each data point. It is controlled by a fraction parameter that dictates the strict percentage of nearby data points considered.

The Forecasting Engine: Deconstructing ARIMA

AR

(Auto-Regressive)

Parameter: p

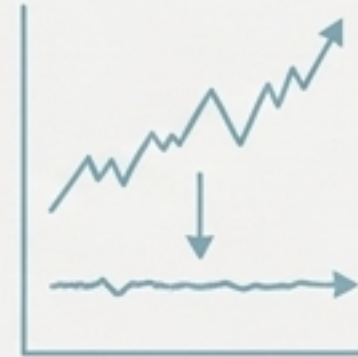


Uses the dependent relationship between an observation and a specific number of lagged observations. Relies mathematically on the PACF (Partial Autocorrelation) plot to determine the value of ' p '.

I

(Integrated)

Parameter: d



The exact number of differences required to strip away trends and make the raw time series stationary. Relies entirely on Unit Root tests (ADF / KPSS) to determine the value of ' d '.

MA

(Moving Average)

Parameter: q



Uses the dependency between an observation and a residual error from a moving average model applied to lagged observations. Relies on the ACF (Autocorrelation) plot to determine the value of ' q '.

Model Declaration: ARIMA (p , d , q)

The Master Blueprint: Time Series Workflow Synthesis

